

1. What is a blockchain?

- a) A physical chain of computer servers
- b) A centralized database owned by a bank
- c) A distributed, immutable ledger
- d) A type of software for creating websites

Answer: c) A distributed, immutable ledger

2. What is a "block" in a blockchain?

- a) A security firewall
- b) A user's personal computer
- c) A collection of transactions added to the chain
- d) A piece of cryptocurrency

Answer: c) A collection of transactions added to the chain

3. What does it mean for a blockchain to be "immutable"?

- a) It is completely anonymous
- b) It is very fast
- c) It is controlled by a single person
- d) Once data is recorded, it cannot be changed or deleted

Answer: d) Once data is recorded, it cannot be changed or deleted

4. What is the very first block in any blockchain called?

- a) The Alpha Block
- b) The Original Block
- c) The Genesis Block
- d) The Prime Block

Answer: c) The Genesis Block

5. What is a "hash"?

- a) A digital signature
- b) A type of cryptocurrency coin
- c) A unique, fixed-length string of characters representing data
- d) A password to access a crypto wallet

Answer: c) A unique, fixed-length string of characters representing data

6. What is "mining" in a Proof-of-Work blockchain like Bitcoin?

- a) Physically digging for valuable minerals
- b) Creating new crypto wallets
- c) Solving complex mathematical puzzles to validate transactions and create a new block
- d) Analyzing market data to predict crypto prices

Answer: c) Solving complex mathematical puzzles to validate transactions and create a new block

7. Which of the following is an example of a public blockchain?

- a) A company's internal supply chain ledger
- b) A private government voting system

- c) Hyperledger Fabric
- d) Bitcoin

Answer: d) Bitcoin

8. What does a cryptocurrency wallet actually store?

- a) Digital coins
- b) Physical cash
- c) Public and private keys
- d) Credit card information

Answer: c) Public and private keys

9. Which key is used to digitally sign a transaction, proving ownership?

- a) The public key
- b) The private key
- c) The master key
- d) The session key

Answer: b) The private key

10. What is the main benefit of a decentralized network?

- a) It is faster than a centralized network
- b) It has a single point of control for easy management
- c) It has no single point of failure and is resistant to censorship
- d) It requires less electricity to operate

Answer: c) It has no single point of failure and is resistant to censorship

11. What is a "smart contract"?

- a) A legally binding paper contract
- b) A computer program that automatically executes when specific conditions are met
- c) A financial agreement between two banks
- d) A security protocol for a crypto wallet

Answer: b) A computer program that automatically executes when specific conditions are met

12. Which blockchain platform is most famous for pioneering the use of smart contracts?

- a) Bitcoin
- b) Ripple
- c) Ethereum
- d) Dogecoin

Answer: c) Ethereum

13. How are blocks linked together to form the "chain"?

- a) Through physical wires
- b) By a central server's index
- c) Each block contains the cryptographic hash of the previous block
- d) They are linked by user passwords

Answer: c) Each block contains the cryptographic hash of the previous block

14. What is a "nonce" in the context of Bitcoin mining?

- a) A user's nickname
- b) The amount of Bitcoin being transacted
- c) A random number that miners must find to solve the block puzzle
- d) A type of fee paid to the network

Answer: c) A random number that miners must find to solve the block puzzle

15. Bitcoin's consensus algorithm is known as:

- a) Proof of Stake (PoS)
- b) Proof of Authority (PoA)
- c) Proof of Work (PoW)
- d) Proof of History (PoH)

Answer: c) Proof of Work (PoW)

16. A "ledger" is a:

- a) Computer programming language
- b) Record-keeping book or file
- c) Type of hardware for mining
- d) Marketing term for a new coin

Answer: b) Record-keeping book or file

17. Which part of a transaction is visible to everyone on a public blockchain?

- a) The sender's and receiver's private keys
- b) The sender's and receiver's real-world identities
- c) The sender's and receiver's public addresses and the transaction amount
- d) The password used to authorize the transaction

Answer: c) The sender's and receiver's public addresses and the transaction amount

18. Blockchain is a type of:

- a) Centralized database
- b) Distributed ledger
- c) Spreadsheet software
- d) Cloud server

Answer: b) Distributed ledger

19. Blockchain is primarily secured through:

- a) Passwords
- b) Encryption
- c) Cryptographic hashing
- d) Firewalls

Answer: c) Cryptographic hashing

20. Each block in the blockchain contains:

- a) Data, previous hash, current hash
- b) Only data
- c) Only timestamp
- d) Only previous block address

Answer: a) Data, previous hash, current hash

21. What does a “node” do in a blockchain network?

- a) Deletes transactions
- b) Mines coins
- c) Maintains a copy of the ledger
- d) Controls user access

Answer: c) Maintains a copy of the ledger

22. Blockchain technology eliminates the need for:

- a) Cryptography
- b) Central intermediaries
- c) Computers
- d) Data verification

Answer: b) Central intermediaries

23. Which of these is NOT a feature of blockchain?

- a) Transparency
- b) Decentralization
- c) Reversibility
- d) Security

Answer: c) Reversibility

24. A blockchain fork occurs when:

- a) Two miners find a block at the same time
- b) A block is deleted
- c) A blockchain stops
- d) A transaction fails

Answer: a) Two miners find a block at the same time

25. In blockchain, data is stored in:

- a) Tables
- b) Sheets
- c) Blocks
- d) Trees

Answer: c) Blocks

26. What is a “Gas Fee” in Ethereum?

- a) Fuel charge
- b) Transaction cost
- c) Network bandwidth
- d) Miner bonus

Answer: b) Transaction cost

27. Which of the following best describes a “block” in blockchain?

- a) A single line of code
- b) A storage unit containing transaction data
- c) A type of token
- d) A miner’s wallet

Answer: b

28. Who created Bitcoin, the first blockchain-based cryptocurrency?

- a) Vitalik Buterin
- b) Satoshi Nakamoto
- c) Charles Hoskinson
- d) Gavin Wood

Answer: b

29. What is the main purpose of using cryptography in blockchain?

- a) To slow down mining
- b) To secure and verify transactions
- c) To increase network size
- d) To delete transaction history

Answer: b

30. Which of these best explains the term “decentralization”?

- a) All data stored on one server
- b) Control shared among multiple nodes instead of a single entity
- c) A faster mining method
- d) Use of multiple currencies

Answer: b